

Skill Week2:

Assignment-1:

```
samanth-ubuntu@samanth-ubuntu:~/OSSP$ mkdir src include obj bin docs tests scripts assets
samanth-ubuntu@samanth-ubuntu:~/OSSP$ ls
assets bin docs include obj scripts src tests
samanth-ubuntu@samanth-ubuntu:~/OSSP$ touch .gitignore README.md LICENSE Makefile
samanth-ubuntu@samanth-ubuntu:~/OSSP$ touch src/main.c src/shell.c src/parser.c src/executor.c src/builtin.c
samanth-ubuntu@samanth-ubuntu:~/OSSP$ touch include/shell.h include/parser.h include/executor.h include/builtin.h
samanth-ubuntu@samanth-ubuntu:~/OSSP$ touch tests/test_parser.c tests/test_executor.c
samanth-ubuntu@samanth-ubuntu:~/OSSP$ touch scripts/run.sh
samanth-ubuntu@samanth-ubuntu:~/OSSP$ touch docs/design.md docs/report.pdf
samanth-ubuntu@samanth-ubuntu:~/OSSP$ touch assets/architecture.png
samanth-ubuntu@samanth-ubuntu:~/OSSP$ tree
Command 'tree' not found, but can be installed with:
sudo snap install tree # version 2.1.3+pkg-5852, or
sudo apt install tree # version 2.3.1-1
See 'snap info tree' for additional versions.
samanth-ubuntu@samanth-ubuntu:~/OSSP$ sudo apt install tree
[sudo: authenticate] Password:
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Installing:
  tree

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 0
  Download size: 52.7 kB
  Space needed: 162 kB / 22.1 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute/universe arm64 tree arm64 2.3.1-1 [52.7 kB]
Fetched 52.7 kB in 1s (52.0 kB/s)
Selecting previously unselected package tree.
(Reading database... 109586 files and directories currently installed.)
Preparing to unpack .../tree_2.3.1-1_arm64.deb...
Unpacking tree (2.3.1-1)...
Setting up tree (2.3.1-1)...
Processing triggers for man-db (2.13.1-1build1)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date (ABI upgrades are not detected).

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
```

```

Fetched 52.7 kB in 1s (52.0 kB/s)
Selecting previously unselected package tree.
(Reading database... 108586 files and directories currently installed.)
Preparing to unpack ./tree_2.3.1-1_arm64.deb...
Unpacking tree (2.3.1-1)...
Setting up tree (2.3.1-1)...
Processing triggers for man-db (2.13.1-1build1)...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date (ABI upgrades are not detected).

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
samanth-ubuntu@samanth-ubuntu:~/OSSP$ tree
.
├── assets
│   └── architecture.png
├── bin
├── docs
│   ├── design.md
│   └── report.pdf
├── include
│   ├── builtin.h
│   ├── executor.h
│   ├── parser.h
│   └── shell.h
├── LICENSE
├── Makefile
├── obj
├── README.md
├── scripts
│   └── run.sh
├── src
│   ├── builtin.c
│   ├── executor.c
│   ├── main.c
│   ├── parser.c
│   └── shell.c
├── tests
│   ├── test_executor.c
│   └── test_parser.c
└── 9 directories, 18 files
samanth-ubuntu@samanth-ubuntu:~/OSSP$ _

```

This assignment creates an organized OSSP project structure containing source files, header files, tests, documentation and scripts. mkdir is used to create directories, touch creates files, and tree displays the complete project structure in tree format.

Assignment 2:

Task-1:

```
samanth-ubuntu@samanth-ubuntu:~/OSSP$ echo $$
8788
samanth-ubuntu@samanth-ubuntu:~/OSSP$ s
s: command not found
samanth-ubuntu@samanth-ubuntu:~/OSSP$ ps
  PID TTY          TIME CMD
  8788 tty1        00:00:00 bash
 23199 tty1        00:00:00 ps
samanth-ubuntu@samanth-ubuntu:~/OSSP$ sleep 100 &
[1] 23212
samanth-ubuntu@samanth-ubuntu:~/OSSP$ ps
  PID TTY          TIME CMD
  8788 tty1        00:00:00 bash
 23212 tty1        00:00:00 sleep
 23222 tty1        00:00:00 ps
samanth-ubuntu@samanth-ubuntu:~/OSSP$ pstree -p $$
bash(8788)─pstree(23232)
            └sleep(23212)
samanth-ubuntu@samanth-ubuntu:~/OSSP$
```

Task-2:

```
samanth-ubuntu@samanth-ubuntu:~/OSSP$ nano process.c
[1]+  Done                  sleep 100
samanth-ubuntu@samanth-ubuntu:~/OSSP$ gcc process.c -o process
samanth-ubuntu@samanth-ubuntu:~/OSSP$ ./process
Enter command: ls
This is the Parent Process.
Parent PID = 3172
Child PID = 3173
This is the Child Process.
Child PID = 3173
LICENSE    README.md  bin       include   process   scripts   tests
Makefile   assets    docs      obj        process.c src
Child Process Completed.
samanth-ubuntu@samanth-ubuntu:~/OSSP$
```

`fork()` creates a child process from the parent process and gives both processes separate process IDs. `exec()` allows the child process to execute the command entered by the user while the parent waits for the child to complete.